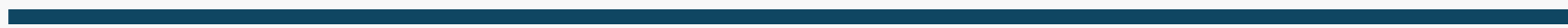


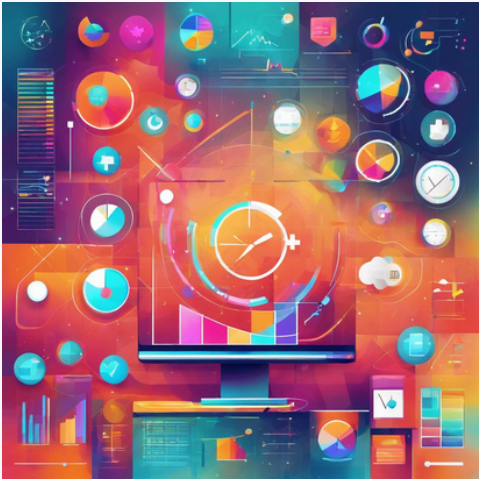
Department of Computer Science
ENGR 202- Data Science and AI Term Project - Fall 2024

Medical Insurance Dataset

29th November 2024



Brief Description about the Medical Insurance Data Set



Dataset:

- The Medical Insurance dataset contains records of individuals' demographic, health, and lifestyle information, along with their insurance charges.

Features & label:

- Features: Age, sex, BMI, children, smoker, and region.
- Label: Charges

Objective:

- Predict medical insurance charges based on the factors.

Dataset Insight:

- Demographic and lifestyle data with target as charges.

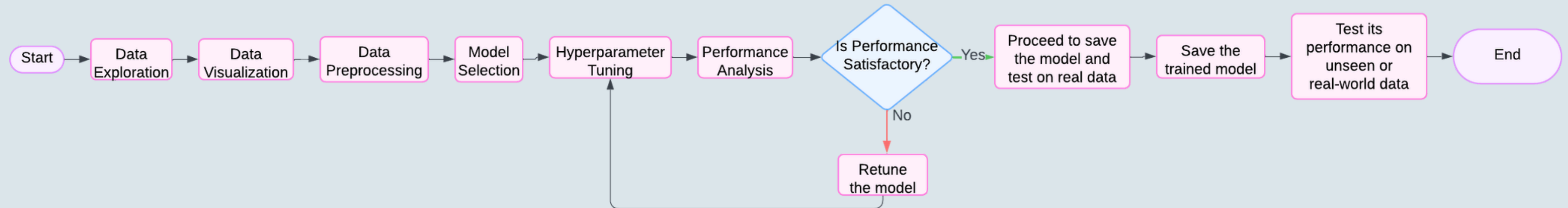
Dataset Statistics:

- The total entries were 1137.

	age	bmi	children	charges
count	1137.000000	1137.000000	1137.000000	1137.000000
mean	35.681618	30.550752	0.985928	13285.245207
std	21.581298	6.124758	1.305384	12068.858512
min	-63.000000	15.960000	-5.000000	1121.873900
25%	25.000000	26.180000	0.000000	4779.602300
50%	38.000000	30.210000	1.000000	9504.310300
75%	51.000000	34.485000	2.000000	16586.497710
max	64.000000	53.130000	5.000000	63770.428010



Workflow Diagram



Data Exploration

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1137 entries, 0 to 1136
Data columns (total 7 columns):
#   Column      Non-Null Count  Dtype
---  -
0   age         1137 non-null   int64
1   sex         1137 non-null   object
2   bmi         1137 non-null   float64
3   children    1137 non-null   int64
4   smoker      1137 non-null   object
5   region      1080 non-null   object
6   charges     1137 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 62.3+ KB
```

All columns has the same number of rows except region column that has 57 missing values (null cells).

	age	bmi	children	charges
count	1137.000000	1137.000000	1137.000000	1137.000000
mean	35.681618	30.550752	0.985928	13285.245207
std	21.581298	6.124758	1.305384	12068.858512
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Negative values & and skewed label

- **Missing Values:** The 'region' feature has 57 missing values.
- **Negative values** in 'age' and 'children' columns.
- Gender column **inconsistencies** (e.g. invalid entries)
- **Outliers:** Compared to other charges, some are noticeably higher (e.g., 63,770).
- Positive correlation between **Age** and **Charges**, and **BMI** and **Charges**.

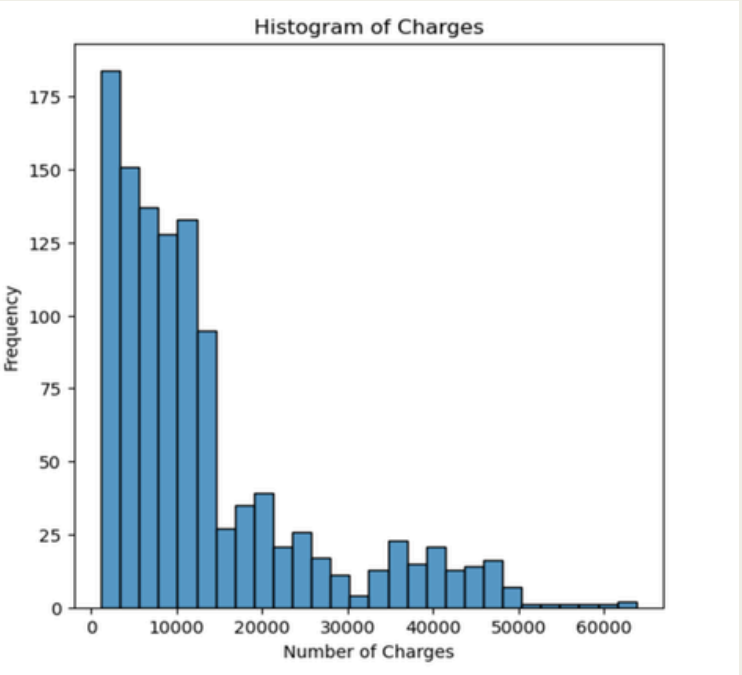
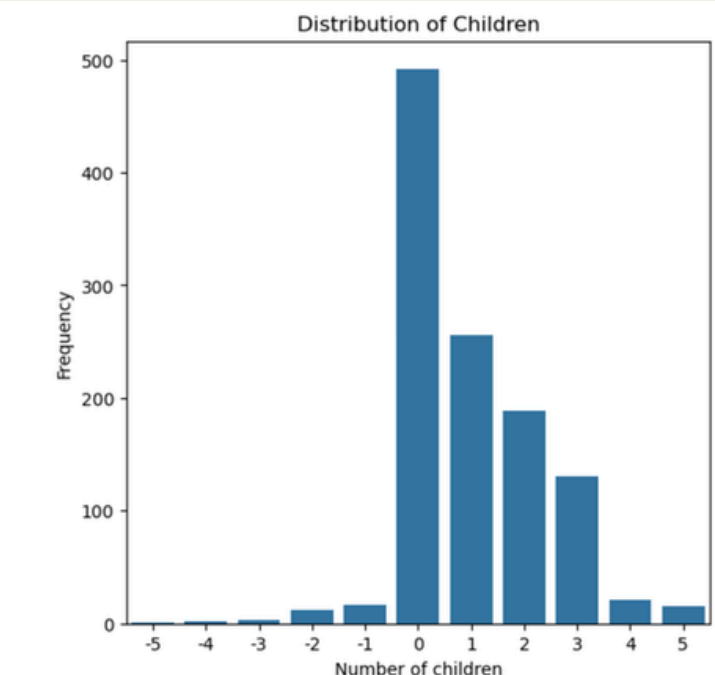
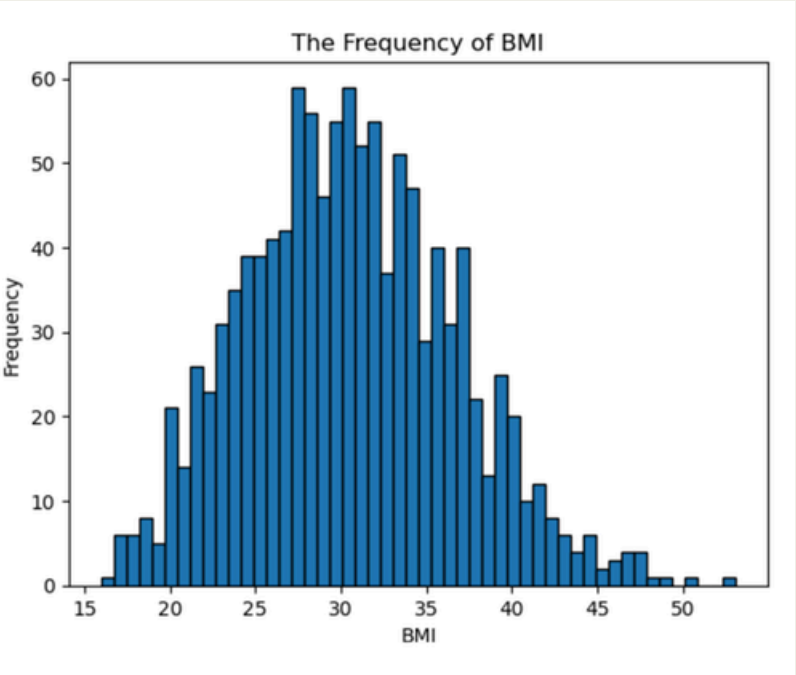
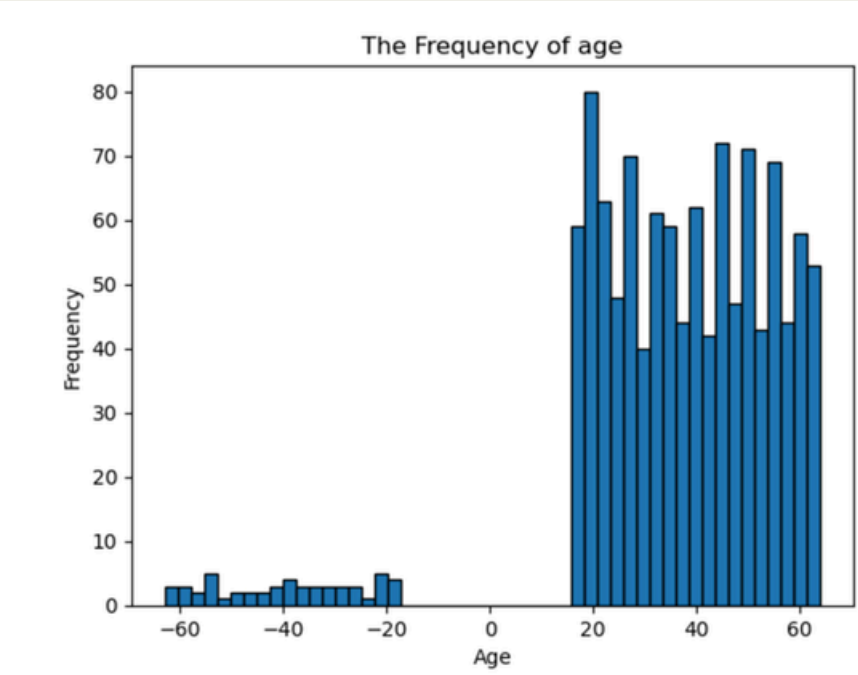
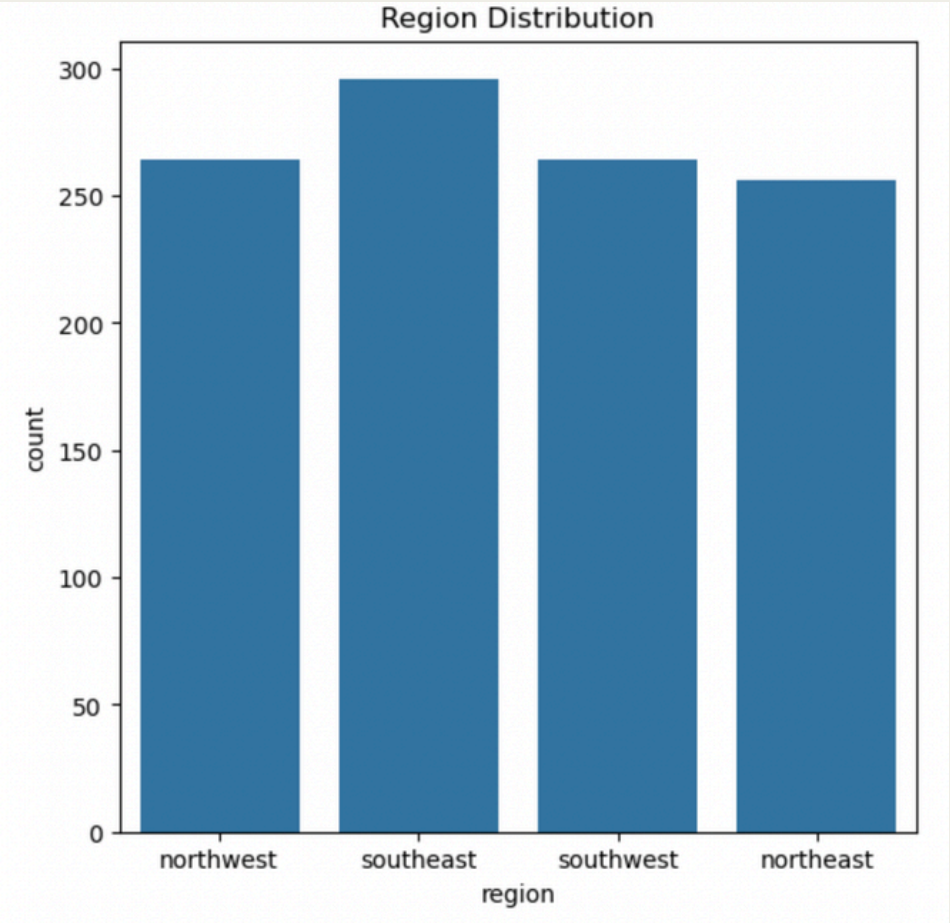
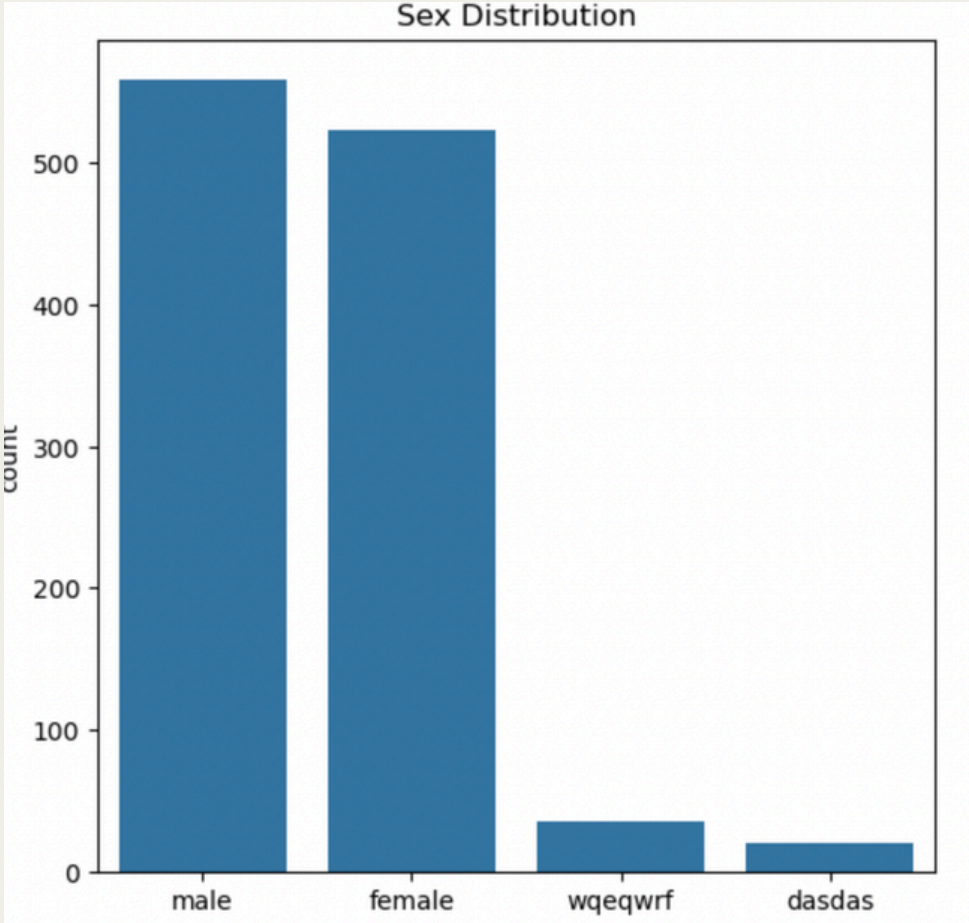
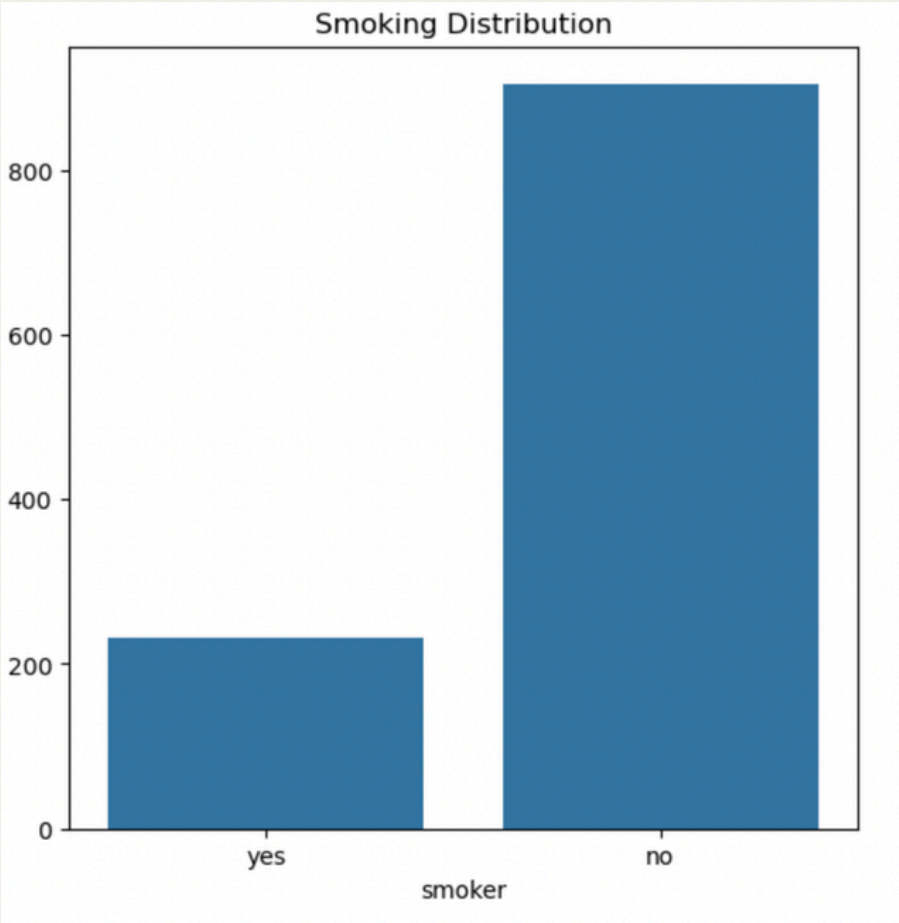
DATA VISUALIZATION

Distribution of genders: Equal numbers of men and women

Status of smoking: Smokers are greatly outnumbered by non-smokers.

Four regions are evenly distributed throughout the territory.

Positive association: Age, BMI, and Charges



Data Preprocessing

1- Splitting Dataset

- 1- Splitting the dataset into features and target variables for prediction tasks.
- 2- Split Dataset into Training (70%), Validation (20%) and Test (10%) Sets.

```
from sklearn.model_selection import train_test_split

# Split the training and test into 70% training and 30% test
X_train, X_test = train_test_split(X_train, test_size=0.3 , random_state=5,shuffle=True)
y_train, y_test = train_test_split(y_train, test_size=0.3 , random_state=5,shuffle=True)

# Split the test into 1/3 for test and 2/3 for validation.
X_val, X_test = train_test_split(X_test, test_size=1/3 , random_state=5,shuffle=True)
y_val, y_test = train_test_split(y_test, test_size=1/3 , random_state=5,shuffle=True)
```

2- Data Cleaning

- 1- Create a function to preprocess training data set, including data cleaning, feature scaling, and encoding of categorical variables.

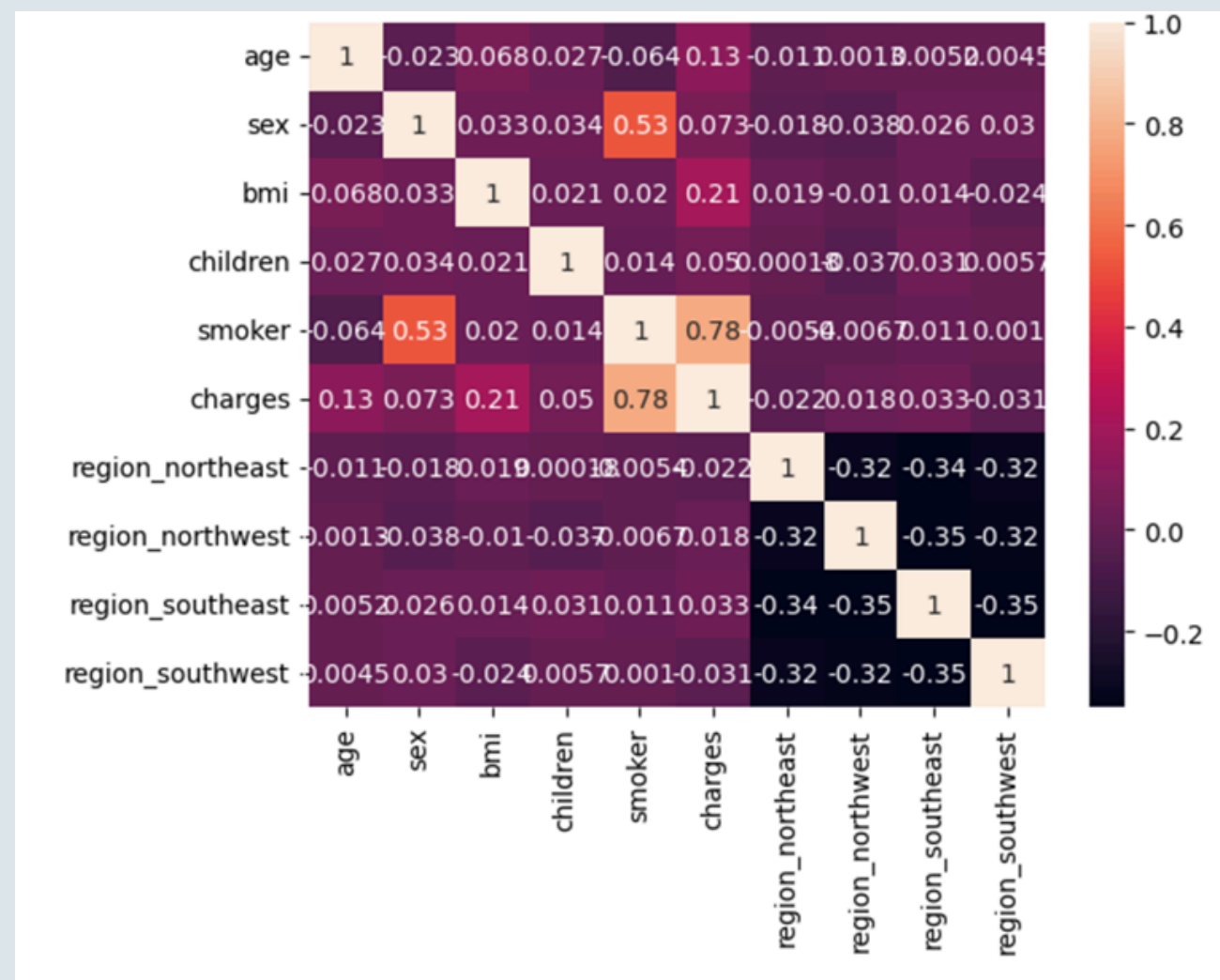
```
def preprocessing_training(training_data, y_train):
```

- 2- Apply the same preprocessing steps we have done on the training data set to the validation data set.

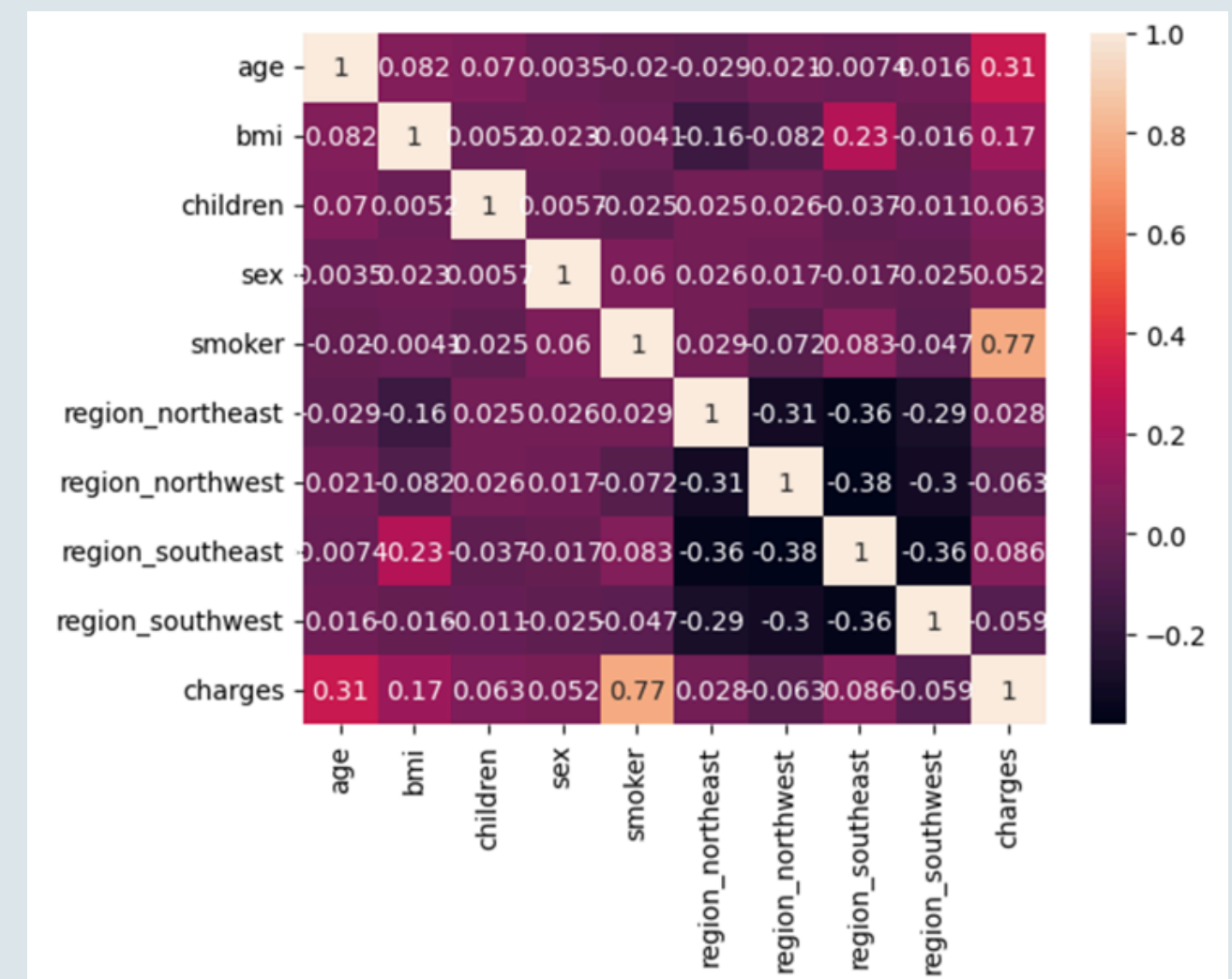
```
def preprocessing_validation(validation_data, y_val, transformer_label_encoder_sex, transformer_label_encoder_smoker,
                             transformer_one_hot_encoder_region, numerical_imputer, categorical_imputer, transformer_scale, medians):
```

Data Preprocessing

- Splitting the dataset into features and target variables for prediction tasks.
- Encoding categorical variables using One-Hot Encoding (e.g., regions) and Label Encoding (e.g., binary features like smoker).
- Cleaning the data by addressing missing values, removing outliers, and correcting invalid entries.
- Scaling numerical data with Min-Max Scaling for normalization.
- Validating the preprocessing with visualizations and analyzing feature correlations with the target variable.



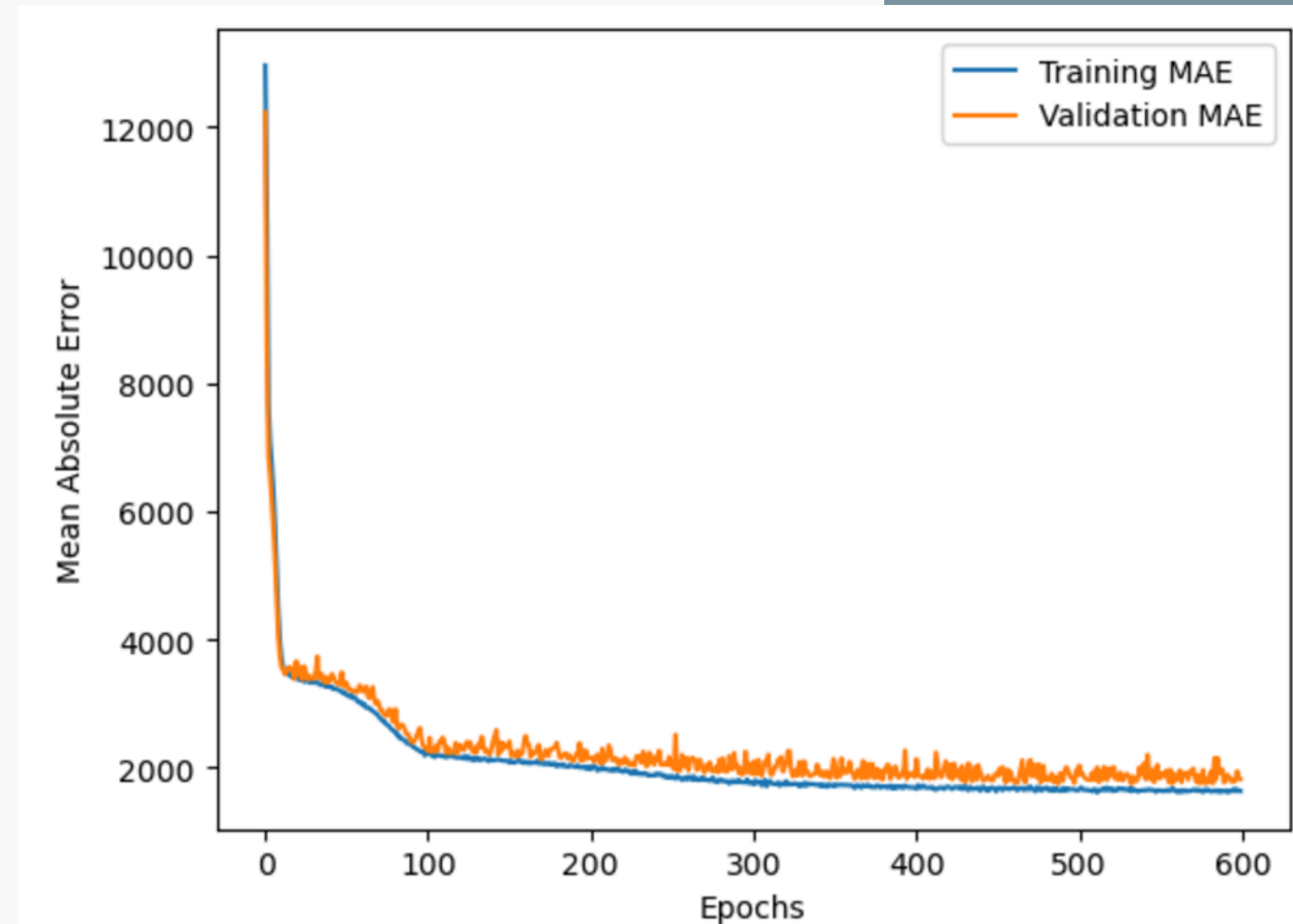
Before Cleaning



After Cleaning

Selected Model

- The model optimized with the RMSprop optimizer was selected.
- Best configuration: model_rmsprop2_600, featuring well-tuned layers, learning rates, and 600 epochs.
- The model showed the best cross-validation performance among all tested configurations.



```
# Plot training and validation MAE
import matplotlib.pyplot as plt

plt.plot(history_rmsprop2_600.history['mae'], label='Training MAE')
plt.plot(history_rmsprop2_600.history['val_mae'], label='Validation MAE')
plt.xlabel('Epochs')
plt.ylabel('Mean Absolute Error')
plt.legend()
plt.show()
```

Hyperparameter Tuning

Initially, we trained the model with 400 epochs, observing a continuous decrease in validation MAE. This indicated that further training could improve performance. Based on this, we increased the epochs to 600, where both training and validation MAE continued to decrease, finally stabilizing. The lower validation MAE at 600 epochs showed better stability, and cross-validation confirmed this improvement, validating our decision.

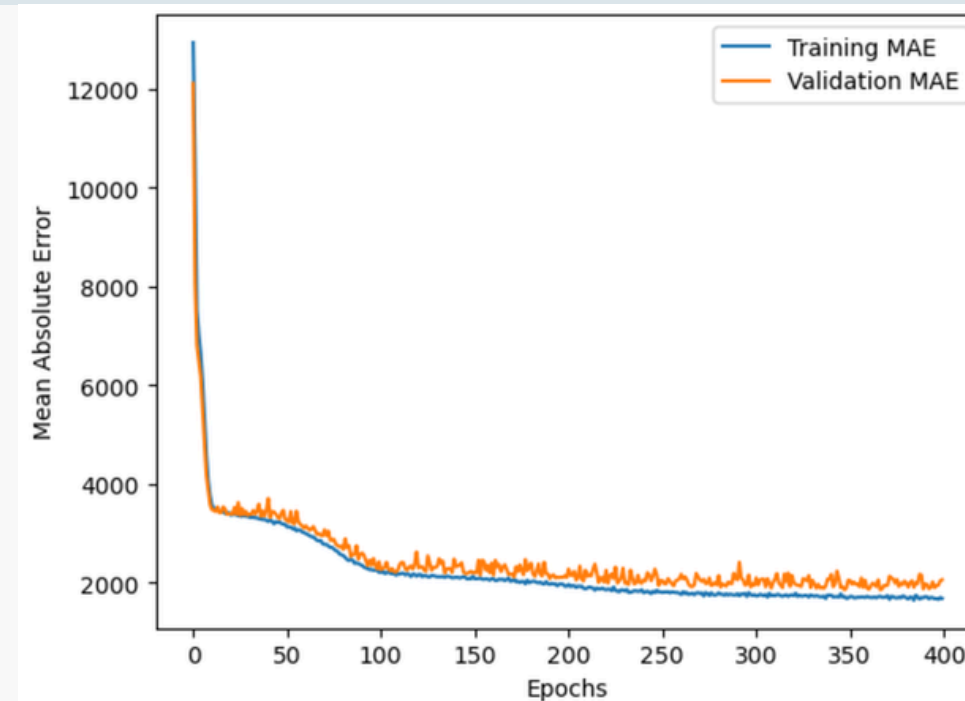
Model: "sequential_6"

Layer (type)	Output Shape	Param #
dense_25 (Dense)	(None, 512)	5,120
dense_26 (Dense)	(None, 256)	131,328
dense_27 (Dense)	(None, 128)	32,896
dense_28 (Dense)	(None, 32)	4,128
dense_29 (Dense)	(None, 1)	33

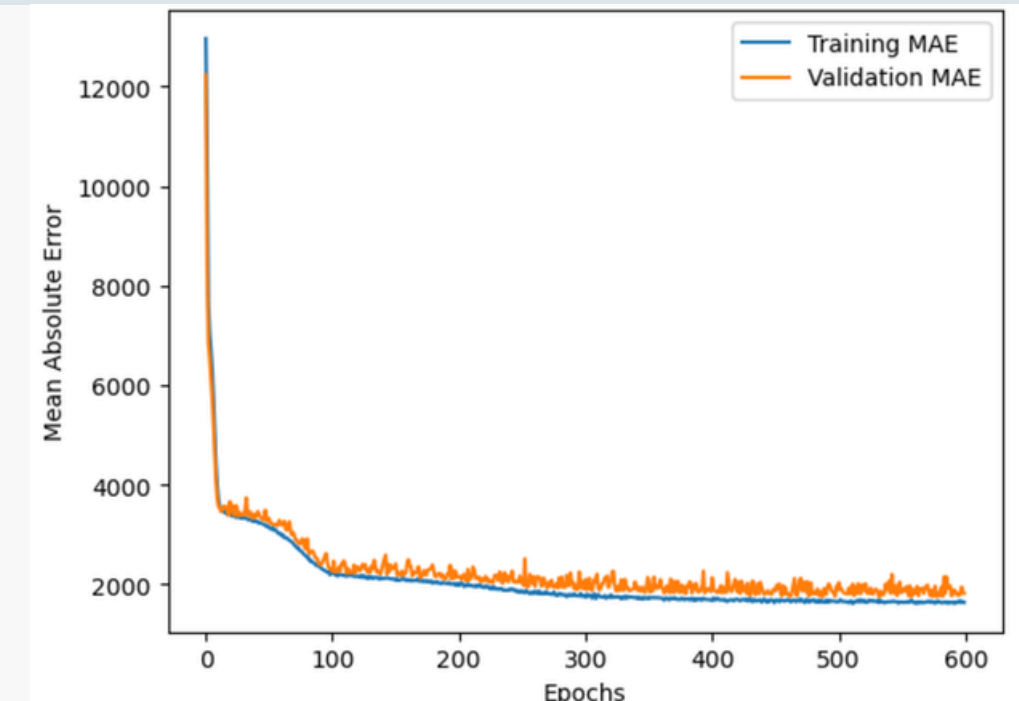
Total params: 173,505 (677.75 KB)

Trainable params: 173,505 (677.75 KB)

Non-trainable params: 0 (0.00 B)

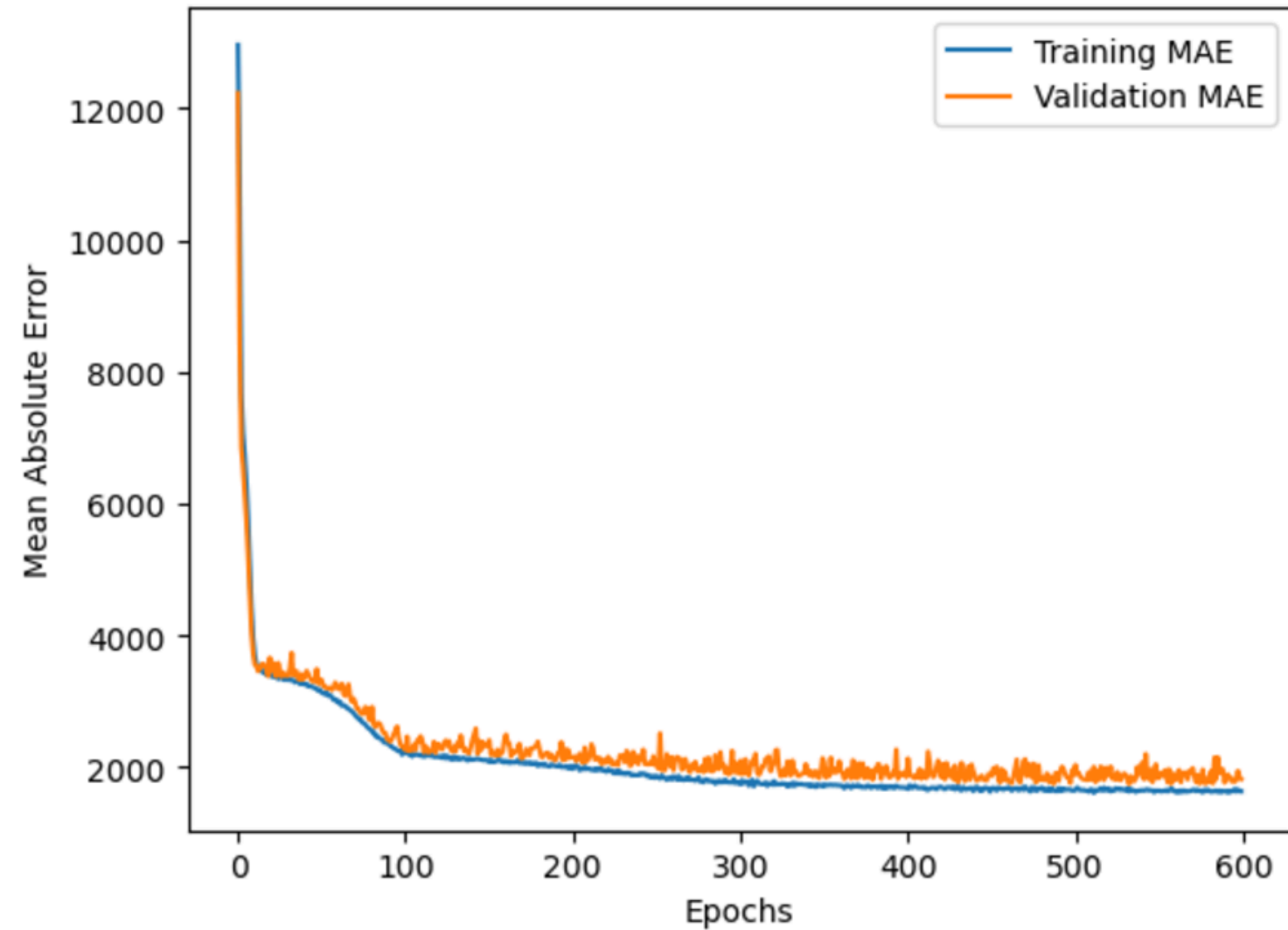


Key features:
Layers : 4
Epochs:400
Batch Size: 32
Optimizer: RMSprop
Dropout layers : No Dropout



Key features:
Layers : 4
Epochs:600
Batch Size: 32
Optimizer: RMSprop
Dropout layers : No Dropout

Model Performance



- Low training and validation loss showed strong learning and accurate results.
- Validation MAE confirmed the model's reliability in predictions.
- Final model saved as **trained_model.keras** for future use (more analysis).

Key features:

Layers : 4

Epochs:600

Batch Size: 32

Optimizer: RMSprop

Dropout layers : No Dropout

